

Messerschmitts. The Hurricanes and Spitfires that were a match for the German fighters soon had to be withdrawn to Britain because there were no airfields left from which they could operate.

Whereas the RAF and the Armée de l'Air had failed to establish any effective system for coordinating air and ground operations, the Luftwaffe was integrated into an overall strategy of shock and mobility. Avoiding sterile disputes about whether or not an air force should be "independent" or subordinate to the other arms, the Luftwaffe used its airplanes sometimes in direct support of army operations

and sometimes on wide-ranging interdiction duties shading into strategic attacks on factories and cities.

Flying artillery

The most striking aspect of the *Blitzkrieg* strategy was the use of aircraft as "flying artillery" in support of armored and motorized forces. On the ground Luftwaffe liaison officers were assigned to panzer units to coordinate air attacks with army operations, while the Luftwaffe's logistical organization kept air units supplied with fuel and munitions as they moved to keep up with the rapidly advancing front line. The Germans showed a keen

appreciation of the psychological factor in warfare, especially the demoralizing effect of air attack on ground troops. The Ju 87 Stuka, the most famous and feared aircraft of the war's *Blitzkrieg* phase, was in some ways a backward airplane for its time, a poorly armed two-seater with a fixed undercarriage. But the wailing soundtrack of its siren in the attack dive struck terror into its victims. Its bombing was accurate enough to destroy communications links such as bridges and railroad junctions, and concentrated in tight formation it could deliver a devastating attack on ground troops or on ships – as during the Battle of Norway in April 1940.

Stukas pilots attacked in formation, rolling over and peeling off into the dive in an accelerating cascade behind the group commander. The Stuka dived at near to vertical, accelerating to over 300mph (480kph) before the air brakes activated to stop it from reaching a velocity that would cause it to break up. Enemy pilots and flak-gunners soon noticed that the Stuka was especially vulnerable pulling out of its dive. At that moment its speed was at its slowest, and the pilot was preoccupied with resetting the machine for level flight. The aircraft was in any case vulnerable to enemy fighters, flying around 100mph (160kph) slower and with poor maneuverability.

As the Luftwaffe lost its ability to guarantee air superiority, the Stuka's usefulness declined.

A general lesson of the first years of the war was the vulnerability of all bombers to attack by fighters. During the Dunkirk evacuation in June 1940, it was not only Stukas but also He 111s and Ju 88s – the latter fast and agile enough to later become a night fighter – that suffered heavily under attack from RAF Spitfires and Hurricanes. It was a lesson confirmed by the Battle of Britain. The single-seat fighter ruled the air (at least by day) and, as in World War I, dogfights pitted fighter against fighter in aerial combat where split-second reaction times determined life or death.

Airborne invasions

In its support of army operations, the Luftwaffe experimented with using aircraft to deliver troops to the point of battle, either by parachute or glider. One of the most successful

GROUND SUPPORT

German ground crew carry out maintenance and refuelling on a Dornier Do 17. The Luftwaffe was superbly organized on the ground to support operations from airstrips on a rapidly moving battlefield.

